higher abstraction

Assembly checked by assembly DRC and LVS

GDS (interposer + chiplets) + assembly-DRC and assembly-LVS reports

IHP Open ADK

the meta-layer: it governs how chiplets are assembled, not how they are designed

Adapters: the bridge

interposer adapter · where chiplet attachment lands

interconnect adapter · method pitch and spacing

the only place a fab layer number is named

Shared registries + schemas

rule_params · interconnect · boundary manifest · layers

one source of truth for every stage

a JSON schema validates each one

ASSEMBLY VERIFICATION HAS TWO AXES, AND NEITHER SUBSUMES THE OTHER

Geometric axis: assembly DRC

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ASM.* placement

overlap · spacing · area · attachment

IXN.* interconnect

spacing · pitch · cross-method

Connectivity axis: assembly LVS

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reference · the declared nets

die pins and the nets that join them

implementation · the placed layouts

pad landings plus interposer routing

Both axes check one placed assembly, described by its .chiplet placement

Ordinary PDKs: the ADK sits one layer above these and never touches their fab rules

Chiplet PDKs

several technologies

dies are verified silicon

Interposer PDK

IHP IntM4TM2

layers, DRC rules

Interconnect PDK

Cu-pillar / micro-bump

methods (pitch, spacing)

A PDK is to a single chip what the ADK is to a multi-chip assembly.